

Homework Section 12.3

1. Let $\mathbf{u} = \langle -2, 5, -7 \rangle$ and $\mathbf{v} = \langle 2, -3, -6 \rangle$. Find the dot product, $\mathbf{u} \cdot \mathbf{v}$.
2. Repeat number 1 for $\mathbf{u} = -\mathbf{i} - 5\mathbf{j} + 7\mathbf{k}$ and $\mathbf{v} = 6\mathbf{i} + 4\mathbf{j} - 3\mathbf{k}$.
3. Find $\mathbf{u} \cdot \mathbf{v}$, given $|\mathbf{u}| = 5$, $|\mathbf{v}| = 11$ and the angle between $\mathbf{u}$ and $\mathbf{v}$ is $\pi/3$.
4. Find the angle between $\mathbf{u} = \langle -2, 5, -7 \rangle$ and $\mathbf{v} = \langle 2, -3, -6 \rangle$. Give an exact expression and also an approximation to the nearest tenth of a degree.
5. Repeat number 4 for the vectors $\mathbf{u} = -\mathbf{i} - 5\mathbf{j} + 7\mathbf{k}$ and $\mathbf{v} = 6\mathbf{i} + 4\mathbf{j} - 3\mathbf{k}$.
6. Decide if the given pairs of vectors are parallel, orthogonal or neither.
 - a) $\mathbf{u} = \langle -2, 5, 2 \rangle$, $\mathbf{v} = \langle 0, -4, 10 \rangle$
 - b) $\mathbf{u} = -\mathbf{i} - 5\mathbf{j} + 7\mathbf{k}$ and $\mathbf{v} = 6\mathbf{i} + 4\mathbf{j} - 3\mathbf{k}$
 - c) $\mathbf{u} = -3\mathbf{i} - 2\mathbf{j} + \frac{3}{2}\mathbf{k}$ and $\mathbf{v} = 6\mathbf{i} + 4\mathbf{j} - 3\mathbf{k}$
7. Find the values of x that make the vectors $\langle 2, x, x \rangle$ and $\langle x, x, -3 \rangle$ orthogonal.
8. Find the vector projection of $\mathbf{u}$ onto $\mathbf{v}$:
 - a) $\mathbf{u} = -\mathbf{i} + 2\mathbf{j} - \mathbf{k}$, $\mathbf{v} = 2\mathbf{i} + \mathbf{j} + \mathbf{k}$
 - b) $\mathbf{u} = \langle -2, 5, -7 \rangle$ and $\mathbf{v} = \langle 2, -3, -6 \rangle$.
9. Find the work done in moving a particle from $P(3, -1, 0)$ to $Q(2, 3, 1)$ if the magnitude and direction of the force is given by $\mathbf{F} = \langle 5, 6, -2 \rangle$. Assume the magnitude of the force is in pounds and the magnitude of $\overrightarrow{PQ}$ is in feet (**Hint:** find the distance vector, $\overrightarrow{PQ}$, in component form first).
10. If $\|\mathbf{F}\|$ is 40 N, $\|\overrightarrow{PQ}\| = 3$ m, and $\theta = 60^\circ$, find the work done by $\mathbf{F}$ in acting from P to Q .
11. The element in the i th row and j th column of a matrix A is denoted a_{ij} . Let $A = \begin{bmatrix} 2 & -3 & 8 \\ -5 & 1 & 0 \end{bmatrix}$.
Which entry is a_{13} ? Which entry is a_{22} ?

12. The **determinant** of a 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is given by $\boxed{\det(A) = ad - bc}$.

Let $A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & -3 \\ 4 & 5 \end{bmatrix}$.

a) Find $\det(A)$

b) Find $\det(B)$

13. Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$. To any entry, a_{ij} , in the matrix, there corresponds a 2×2 sub-matrix

obtained by deleting the i th row and j th column of A . The following table gives the sub-matrices for entries in the first row of A :

Entry	Corresponding Sub-matrix
a_{11}	$\begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}$
a_{12}	$\begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix}$
a_{13}	$\begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$

The **determinant** of a 3×3 matrix can be defined using the above sub-matrices as in the following formula:

$$\boxed{\det(A) = a_{11} \det \begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix} - a_{12} \det \begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix} + a_{13} \det \begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}}$$

Find the following determinants using the above formula:

a) $\begin{bmatrix} -1 & 5 & 2 \\ 0 & 2 & 4 \\ 3 & -5 & 1 \end{bmatrix}$

b) $\begin{bmatrix} 2 & 1 & 0 \\ -4 & 3 & 8 \\ -3 & 1 & 1 \end{bmatrix}$